Course: Foundations of AI with Embeddings

Lecture 6: Sequential Models

Markov Chains · Stationary Distributions · Hidden Markov Models

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Sequences are everywhere

Many real-world systems generate *sequential* data where order matters:

- **Text**: word follows word; character follows character
- **User sessions**: page $\rightarrow$ page $\rightarrow$ purchase (or exit)
- **Finance**: stock prices, trading volumes over time
- **Speech**: a stream of audio frames encoding phonemes
- **Biology**: DNA base sequences, protein residues

Goal: learn a probability distribution over sequences $X_1, X_2, \ldots, X_T$.

Problem: $P(X_1, \ldots, X_T) = P(X_1) \cdot P(X_2|X_1) \cdot P(X_3|X_2, X_1) \cdot \ldots =$ $P(X_1) \prod_{t=2}^{T} P(X_t \mid X_{t-1}, \ldots, X_1)$ is intractable in general — the conditioning history grows without bound.

Key idea: impose a *conditional independence* assumption to tame the complexity.

1. Markov Chains

The Markov Property

A sequence $X_1, X_2, \ldots$ satisfies the **(first-order) Markov property** if

$$\boxed{P(X_t \mid X_{t-1}, X_{t-2}, \ldots, X_1) = P(X_t \mid X_{t-1}).}$$

Intuition

"The future is independent of the past, *given* the present."

Consequence: the joint probability factorises as

$$P(X_1, \ldots, X_T) = P(X_1) \prod_{t=2}^{T} P(X_t \mid X_{t-1}).$$

We additionally assume **time-homogeneity**: $P(X_t = j \mid X_{t-1} = i)$ does not depend on t.

Formal Definition

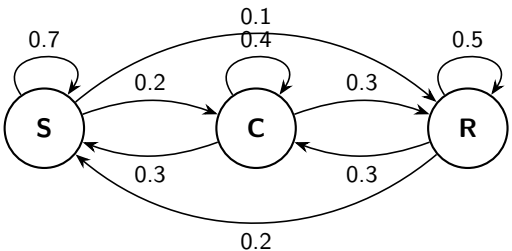
A **(homogeneous, discrete-time) Markov chain** is defined by:

- A finite **state space** $\mathcal{S} = \{s_1, \ldots, s_K\}$
- A **transition matrix** $A \in \mathbb{R}^{K \times K}, \quad A_{ij} = P(X_t = s_j \mid X_{t-1} = s_i)$
- An **initial distribution** $\pi_0 \in \mathbb{R}^K, \quad (\pi_0)_i = P(X_1 = s_i)$

Properties of A:

- All entries non-negative: $A_{ij} \geq 0$
- **Row-stochastic**: $\sum_{j=1}^{K} A_{ij} = 1$ for every row i
- Entry A_{ij} is the probability of moving from state i to state j in one step

Transition Diagram



States: **S**unny, **C**loudy, **R**ainy.

Transition matrix:

$$A = \begin{pmatrix} 0.7 & 0.2 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$$

Row i = distribution over next states when currently in state i.

State Distribution Evolution

Let $\pi_t \in \mathbb{R}^K$ be the **row vector** of state probabilities at time t:

$$(\pi_t)_j = P(X_t = s_j).$$

One-step update (law of total probability):

$$(\pi_t)_j = \sum_{i=1}^{K} (\pi_{t-1})_i \, A_{ij} \qquad \Longrightarrow \qquad \pi_t = \pi_{t-1} A.$$

Multi-step propagation:

$$\pi_t = \pi_0 \, A^t.$$

The (i, j) entry of A^t gives the *t-step transition probability* $P(X_t = s_j \mid X_0 = s_i)$.

Example. Start in Sunny with probability 1: $\pi_0 = (1, 0, 0)$. After one step: $\pi_1 = (0.7,\ 0.2,\ 0.1)$. After two steps: $\pi_2 = \pi_1 A$.

Stationary Distribution

A distribution $\pi^* \in \mathbb{R}^K$ is **stationary** (or *invariant*) for A if

$$\pi^* = \pi^* A, \qquad \sum_{j=1}^{K} \pi_j^* = 1, \qquad \pi_j^* \geq 0.$$

π^* is a *left eigenvector* of A with eigenvalue 1: $\pi^*(A - I) = \mathbf{0}$.

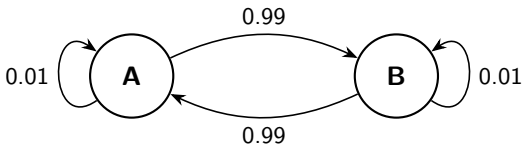
Existence and uniqueness (Perron–Frobenius)

A Markov chain that is **irreducible** (any state reachable from any state) and **aperiodic** (no periodic cycles) has a **unique** stationary distribution, and $\pi_t \to \pi^*$ for any starting distribution π_0.

Computing π^*:

- Solve the linear system $\pi^*(A - I) = \mathbf{0}$ subject to $\sum_j \pi_j^* = 1$
- *Power iteration*: run $\pi_0 A^t$ until convergence

Question: What is the stationary distribution?



Transition matrix:

$$A = \begin{pmatrix} 0.01 & 0.99 \\ 0.99 & 0.01 \end{pmatrix}$$

The chain bounces between A and B almost every step.

What is π^*?

Answer: Stationary Distribution of the Two-State Chain

Setup: $\pi^* = (\pi_A^*,\ \pi_B^*)$ satisfies $\pi^* = \pi^* A$ and $\pi_A^* + \pi_B^* = 1$.

Equation for π_A^*:

$$\pi_A^* = 0.01\,\pi_A^* + 0.99\,\pi_B^* = 0.01\,\pi_A^* + 0.99\,(1 - \pi_A^*)$$

$$\pi_A^* = 0.99 - 0.98\,\pi_A^* \quad \Longrightarrow \quad 1.98\,\pi_A^* = 0.99 \quad \Longrightarrow \quad \boxed{\pi_A^* = \pi_B^* = 0.5}$$

Intuition. The chain is *symmetric*: $a_{AB} = a_{BA} = 0.99$. By symmetry, both states must be equally likely in steady state — *regardless* of how large or small the self-loop probability is.

Key takeaway. The stationary distribution depends on the *ratio* of transition rates, not their absolute magnitude.

General two-state formula

For $A = \left(\begin{smallmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{smallmatrix}\right)$: $\pi_A^* = \dfrac{\beta}{\alpha+\beta}$, $\pi_B^* = \dfrac{\alpha}{\alpha+\beta}$. Here $\alpha = \beta = 0.99$, so $\pi^* = (0.5,\ 0.5)$.

Application: Google PageRank

Naive idea: rank page j by the number of incoming links.

Problem: 100 links from spam sites $<<$ 1 link from Wikipedia.

Recursive definition. $L = (L_{ij}) = \mathbb{1}_{j \to i}$ – a matrix of links, $c_j = \sum_i L_{ij}$ – the number of pages that page j links to. PageRank p_j is the weighted sum of ranks of pages that refer to j:

$$p_j = \sum_{i=1}^{N} \frac{L_{ij}}{c_i} p_i \quad \text{or in matrix form: } p = LD_c^{-1}p,$$

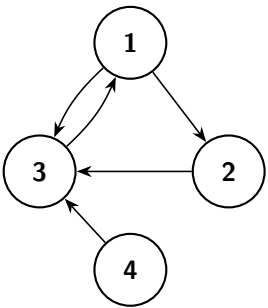
where $D_c = \text{diag}(c)$. So each page i distributes its rank equally among all pages it links to.

Problem: *Dangling nodes* – pages with no outgoing links.
Fix – teleportation (Page, 1998):

$$p = (1-d)e + d \cdot LD_c^{-1}p \qquad d \approx 0.85,\ e = (1, \ldots, 1)^\top$$

$$p = \left[(1-d)ee^\top/N + dLD_c^{-1}\right] p = Ap$$

PageRank: example



Out-degrees: $c = (2,\ 1,\ 1,\ 1)$

$$L = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad LD_c^{-1} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 0.0375 & 0.0375 & 0.8875 & 0.0375 \\ 0.4625 & 0.0375 & 0.0375 & 0.0375 \\ 0.4625 & 0.8875 & 0.0375 & 0.8875 \\ 0.0375 & 0.0375 & 0.0375 & 0.0375 \end{pmatrix}$$

$$p = (1.49, 0.78, 1.58, 0.15)$$

2. Hidden Markov Models

Observed vs. Hidden Sequences

In many settings the underlying *state* cannot be observed directly — only noisy *emissions* are visible:

Application	**Hidden state**	**Observation**
Speech recognition	Phoneme	Audio frame (MFCC vector)
POS tagging	POS tag (Noun, Verb…)	Word
Gene finding	Gene / non-gene region	DNA base (A, C, G, T)
Finance	Market regime (bull/bear)	Daily return
User modelling	User intent	Clicked item

Hidden Markov Model (HMM): extend a Markov chain by adding a layer of *probabilistic emissions* on top of the hidden states.

HMM: Definition

A **Hidden Markov Model** $\lambda = (A, B, \pi)$ consists of:

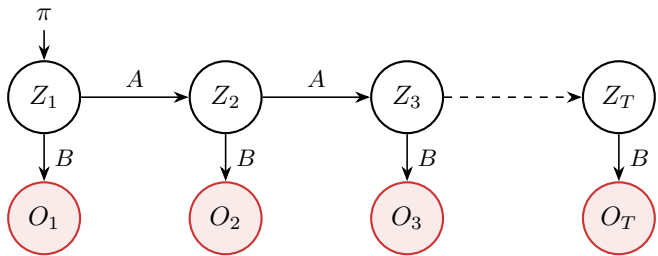
- N **hidden states** $\mathcal{Q} = \{q_1, \ldots, q_N\}$
- M **observation symbols** $\mathcal{V} = \{v_1, \ldots, v_M\}$
- **Transition matrix** $A \in \mathbb{R}^{N \times N}$: $a_{ij} = P(Z_t = q_j \mid Z_{t-1} = q_i)$
- **Emission matrix** $B \in \mathbb{R}^{N \times M}$: $b_{ik} = P(O_t = v_k \mid Z_t = q_i)$
- **Initial distribution** $\pi \in \mathbb{R}^N$: $\pi_i = P(Z_1 = q_i)$

Generative process:

1. Draw $Z_1 \sim \pi$; emit $O_1 \sim B[Z_1, \cdot]$
2. For $t = 2, \ldots, T$: draw $Z_t \sim A[Z_{t-1}, \cdot]$; emit $O_t \sim B[Z_t, \cdot]$

Hidden states $Z_{1:T}$ are *never* observed; we only see $O_{1:T}$.

HMM Structure



- The hidden states $Z_1, \ldots, Z_T$ form a *Markov chain* (governed by A and π)
- Each observation O_t depends *only* on the current hidden state Z_t (governed by B)

Three Fundamental Problems

Given model $\lambda = (A, B, \pi)$ and observations $O = (O_1, \ldots, O_T)$:

1. **Evaluation** — What is the probability of the observed sequence?

$$P(O \mid \lambda) = ?$$

Algorithm: Forward algorithm — Complexity: $O(N^2 T)$

2. **Decoding** — What is the most likely hidden state sequence?

$$Z^* = \arg\max_{Z_{1:T}} P(Z_{1:T} \mid O, \lambda)$$

Algorithm: Viterbi algorithm — Complexity: $O(N^2 T)$

3. **Learning** — Given observations only, how do we estimate λ?

$$\hat{\lambda} = \arg\max_{\lambda} \; P(O \mid \lambda)$$

Algorithm: Baum–Welch (EM) — Complexity: $O(N^2 T)$ per iteration

Problem 1: Evaluation — Forward Algorithm

Naïve approach: sum over all N^T state sequences — exponential in T.

Forward variable: $\alpha_t(i) = P(O_1, \ldots, O_t,\ Z_t = q_i \mid \lambda)$

Initialization $(t = 1)$:

$$\alpha_1(i) = \pi_i \, b_{i,\, O_1}$$

Recursion $(t = 2, \ldots, T)$:

$$\alpha_t(j) = \left[\sum_{i=1}^{N} \alpha_{t-1}(i)\ a_{ij} \right] b_{j,\, O_t}$$

Termination:

$$P(O \mid \lambda) = \sum_{i=1}^{N} \alpha_T(i)$$

The recursion reuses all intermediate computations: $O(N^2 T)$ instead of $O(N^T)$.

Problem 2: Decoding — Viterbi Algorithm

Find the single best path: $Z^* = \arg\max_{Z_{1:T}} P(Z_{1:T} \mid O, \lambda)$.

Viterbi variable: $\delta_t(i) = \max_{Z_{1:t-1}} P(Z_{1:t-1}, Z_t = q_i \mid O_{1:t}, \lambda)$

Initialization:

$$\delta_1(i) = \pi_i \, b_{i,\,O_1}, \qquad \psi_1(i) = 0$$

Recursion:

$$\delta_t(j) = \max_{1 \le i \le N} \big[\delta_{t-1}(i)\; a_{ij}\big]\; b_{j,\,O_t}, \qquad \psi_t(j) = \arg\max_{1 \le i \le N} \big[\delta_{t-1}(i)\; a_{ij}\big]$$

Traceback: start from $Z_T^* = \arg\max_i \; \delta_T(i)$, then $Z_t^* = \psi_{t+1}(Z_{t+1}^*)$ backwards.

Same structure as the forward algorithm with **max** replacing **sum**.

Problem 3: Learning — Baum–Welch (EM)

Goal: find $\lambda = (A, B, \pi)$ maximising $P(O \mid \lambda)$. No closed form — use EM.

E-step: compute soft state assignments via the *forward* (α) and *backward* (β) variables, where

$$\alpha_t(i) = P(O_1, \ldots, O_t, Z_t = q_i, \mid \lambda), \quad \alpha_1(i) = \pi_i \, b_{i,O_1}, \quad \alpha_{t+1}(i) = \sum_j \alpha_t(j) a_{ji} \, b_{i,O_{t+1}}.$$

$$\beta_t(i) = P(O_{t+1}, \ldots, O_T \mid Z_t = q_i, \lambda), \quad \beta_T(i) = 1, \quad \beta_t(i) = \sum_j a_{ij} \, b_{j,O_{t+1}} \, \beta_{t+1}(j).$$

Problem: It works, but it suffers from converging exponentially to zero for long sequences.

Problem 3: Learning — Baum–Welch (EM)

E-step:

$$\gamma_t(i) = P(Z_t = q_i \mid O, \lambda) = \frac{P(Z_t = q_i, O \mid \lambda)}{P(O \mid \lambda)} \propto \alpha_t(i)\beta_t(i)$$

$$\xi_t(i,j) = P(Z_t = q_i, Z_{t+1} = q_j \mid O, \lambda) = \frac{P(Z_t = q_i, Z_{t+1} = q_j, O \mid \lambda)}{P(O \mid \lambda)} \propto \alpha_t(i)\, a_{ij}\, b_{j,O_{t+1}}\, \beta_{t+1}(j)$$

M-step: re-estimate parameters:

$$\hat{\pi}_i = \gamma_1(i), \qquad \hat{a}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}, \qquad \hat{b}_{ik} = \frac{\sum_{t:\, O_t = v_k} \gamma_t(i)}{\sum_{t=1}^{T} \gamma_t(i)}$$

Iterate E and M steps until convergence (guaranteed to find a local maximum of $P(O \mid \lambda)$).

Example: Part-of-Speech Tagging

Task: assign a POS tag to each word in a sentence.

"The cat sat on the mat" $\xrightarrow{\text{Viterbi}}$ Det Noun Verb Prep Det Noun

Model components

- *Hidden states*: POS tags
- *Observations*: words (vocabulary)
- *Transitions*: $P(\text{Noun} \mid \text{Det})$
- *Emissions*: $P(\text{“cat”} \mid \text{Noun})$

Mapping to the three problems

- *Evaluation*: how likely is this sentence?
- *Decoding*: what POS sequence best explains the words? (Viterbi)
- *Learning*: supervised (annotated corpora) or unsupervised (Baum–Welch)

References

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